Discussion Document

CS 271P · Artificial Intelligence · Fall 2026

One per group. Fill it in as you go, and submit it on Gradescope before you leave.

Everywhere you see a grey prompt like *Type here*, click it and replace it with your own words. Type as much as you need. Every answer area grows to fit what you write.

Fill this in first

Programming assignment	*PA3*
Group number	*21*
Date	*2026-10-09*
TA	*Jackie Xin*
Group members	ISMAIL, ROSHAN AGUIRRE, SAMUEL BENJAMIN, PHAN, HUY GIA

Step 1. On your own

Take about five minutes. Before you talk to anyone, write how *you* would approach the problem. A rough idea is enough. Every member writes their own row.

Name	Your first thoughts
AGUIRRE, SAMUEL BENJAMIN	*One approach we can use is alpha-beta pruning and using one or two heuristic. Which could be the two end conditions stated in the problem.*
ISMAIL, ROSHAN	*We run multiple simulations of the game in order to gather information and select the node with the best wi value. We also choose our exploration parameter c to be 0.5 since its the optimal standard for Los Alamos Chess.*

Name	Your first thoughts
PHAN, HUY GIA	*We only allow the agent to go at most 4-5 move ahead(deeper layer), then use some heuristic to estimate how good this node is, and compute UTC value, choosing moves based on UTC*
Your name	*How would you approach this? A rough idea is fine.*

A group of three leaves the last row empty.

Step 2. With your group

Share what each of you wrote. Write the approach your group has settled on. If it builds on one member's idea, say what you changed and why.

- The approach our group came to is using random play outs to gather some initial information, while also using some sort of heuristic to determine. We would only go down n number of moves during our initial play-out. After this we would use the UCT formula to either explore or exploit certain paths in the game, while keeping the end states in mind.

Step 3. With the other group

Meet the group you are paired with and trade approaches. Write anything new you learned, and any change you would now make to your own approach.

- The other group decided to use a similar approach, however, we learned that choosing the c term for UCT is not trivial in some sense.

Step 4. The approach your group will take

Write the approach you intend to use. Be concrete enough that someone could start coding from it. This is the text one of you posts on Ed Discussion.

- After consulting the internet, we determined that using a c value of 0.5 is optimal. However, more information is needed to confirm how this c value should be selected.

Before you leave

Do both of these.

1. **Submit this document on Gradescope.** One submission per group.

2. **Post your Step 4 approach on Ed Discussion.**

Posted on Ed Discussion by	*Samuel Aguirre*

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